

Mahdi Hussain

British Citizen | London | 07514737839 | mhussain18m@gmail.com | Portfolio: <https://mahdi-hussain-portfolio.netlify.app>

Skills: TypeScript, JavaScript (ES6+), Java, Python, React.js, Node.js, Express, Spring Boot, RESTful APIs, PostgreSQL, MySQL, Git, GitHub Actions, CI/CD workflows, Bash, Linux, HTML5, CSS3, Object-Oriented Programming, Agile

EDUCATION

BSc Computer Science - First Class Honours (1:1, 78+%) | University of Leicester

July 2025

Notable Modules: Algorithms, Data Structures and Advanced Programming; Object Oriented Programming; Software Architecture and System Development; Databases and Domain Modelling; User Interface Design Evaluation, Internet and Cloud, Mobile and Web Development

WORK EXPERIENCE

Freelance Software Engineer | Purley Wasl | <https://purleywasl.co.uk>

Feb 2026 – Present

- Leading development of a digital community platform for businesses and services in Purley, translating stakeholder requirements into scalable, production-ready features
- Architected a full-stack system using Python (FastAPI), PostgreSQL (SQLAlchemy), and a TypeScript, React, Vite frontend
- Applied test-driven development practices using pytest and HTTP client testing to ensure reliability of backend services

PROJECTS

AI-Powered Bank Statement Analyser | <https://spend-bee.netlify.app>

- Developed a full-stack AI-driven platform using React, TypeScript, Node.js, and Express, integrating OpenAI to automate transaction categorisation and generate actionable financial insights
- Designed production-ready CI/CD workflows with GitHub Actions to automate build validation and ESLint checks, deploying continuously to Netlify and Railway to enforce consistent release quality
- Built headless end-to-end tests with Playwright as part of the CI/CD pipeline to validate functionality and prevent regressions

Merchant Management System | <https://github.com/mh1803/Merchant-Management-System>

- Built a production-style backend CRM for merchant onboarding, KYB verification, and lifecycle management using Node.js, Express, PostgreSQL, JWT, and Zod, delivering the system in 72 hours using Codex to accelerate development
- Ensured system reliability through comprehensive Jest (service-layer) and Supertest (HTTP) testing, and improved developer efficiency by building bash-based CLI tooling for operational workflows

Event Booking Platform | <https://eventqibla.netlify.app>

- Developed a full-featured event booking platform using TypeScript, React, Node.js, and Express for event creation and discovery
- Reduced frontend render times to under 1 second through lazy loading and memoization, improving application performance
- Optimised RESTful API handling with Node.js and Express, improving code maintainability and data flow
- Enhanced PostgreSQL query performance through strategic indexing, improving response times for high-traffic endpoints

IBM SkillsBuild Companion App

- Commissioned by IBM to develop a gamified web application, successfully increasing user activity by 30% in a mock trial
- Built backend services using Java and Spring Boot, implementing RESTful APIs for business logic and data management
- Configured a relational database with MySQL for data storage and role-based access to support structured user management
- Spearheaded the Agile development process as Scrum Master for a team of 8, managing weekly standups and sprint events for timely completion of project milestones
- Utilised GitLab for documentation and version control, fostering collaboration and maintaining code integrity

Private Cloud Whiteboard

- Distributed a shared, interactive whiteboard application across multiple Microsoft Azure App Service nodes
- Achieved persistent, low-latency connections by integrating Azure Web PubSub for WebSocket communication and Redis Cache for state synchronisation and rapid data retrieval
- Applied Docker containers to simplify horizontal scaling, allowing quick instance deployment during peak demand

London Landmark Finder

- Developed a web application in HTML, CSS, and JavaScript that assists users in discovering landmarks across London
- Secured user sessions with JSON Web Tokens (JWT), enhancing authentication reliability and session management